

API Reference

Cheap Public Transport API

API Version: 0.1.0

This API find the cheapest journey between UK cities via public transport. This combines national rail services, FlixBus coaches, and National Express coaches.

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Security and Authentication

SECURITY SCHEMES

KEY	TYPE	DESCRIPTION
HTTPBearer	http, bearer	

API

1. AUTHENTICATION

Endpoints to authenticate users via Supabase.

1.1 POST /auth/signup

Signup

Register a new account

Creates a new user record in the system with a default 'user' role.

Once registered, the user can authenticate via the login endpoint to receive a JWT access token.

REQUEST

REQUEST BODY - application/json

```
{
  email*    string
  password* string
}
```

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
{
  message* string
  id*      string
}
```

STATUS CODE - 400: Bad Request - Email already registered or password does not meet requirements.

STATUS CODE - 422: Validation Error - Improperly formatted request body.

1.2 POST /auth/login

Login

Authenticate user

Verifies user credentials and returns a set of JWT tokens for subsequent requests.

The access_token must be provided in the 'Authorization' header as a 'Bearer' token for all protected endpoints. The refresh_token is used to obtain a new access token once the current one has expired.

REQUEST

REQUEST BODY - application/json

```
{
  email*    string
  password* string
}
```

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
{
  access_token*  string
  refresh_token* string
  token_type     string DEFAULT:bearer
}
```

STATUS CODE - 401: Unauthorized - Invalid email or password credentials.

STATUS CODE - 422: Validation Error - Improperly formatted request body.

2. CITIES

The 200 largest UK cities with latitude and longitude.

2.1 GET /city/

Get All Cities

Get all cities

Retrieves a complete list of every city currently stored in the database.

Access Level: Admin and User.

REQUEST

No request parameters

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
[ {  
  Array of object:  
    name*      string  
    latitude*   string  
    longitude*  string  
    id*         integer  
  } ]
```

STATUS CODE - 401: Unauthorized - Missing or invalid authentication token.

STATUS CODE - 403: Forbidden - User role not recognized.

2.2 POST /city/

Create New City

Create a new city

Adds a unique city record to the database including coordinates for mapping.

Access Level: Admin only.

Notes:

- City names are automatically converted to lowercase before saving.
- The name field must be unique; duplicate entries will result in an error.

REQUEST

REQUEST BODY - application/json

```
{  
  name*      string  
  latitude*  
    ANY OF  
  prop0  
    number  
    between -90 and 90
```

```

    prop1
    string
    PATTERN:^(?!^[-+.*$][+-]?0*\d*\.\d*$
longitude*
    ANY OF
    prop0
    number
    between -180 and 180
    prop1
    string
    PATTERN:^(?!^[-+.*$][+-]?0*\d*\.\d*$
}

```

RESPONSE

STATUS CODE - 201: Successful Response

RESPONSE MODEL - application/json

```

{
  name*      string
  latitude*  string
  longitude* string
  id*        integer
}

```

STATUS CODE - 401: Unauthorized - Missing or invalid authentication token.

STATUS CODE - 403: Forbidden - User does not have administrative privileges.

STATUS CODE - 422: Validation Error - Input data is invalid or name is not unique.

2.3 GET /city/id/{city_id}

Get City By Id

Get city by ID

Fetches a specific city's details using its unique integer ID.

Access Level: Admin and User.

REQUEST

PATH PARAMETERS

NAME	TYPE	DESCRIPTION
*city_id	integer	

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```

{
  name*      string
  latitude*  string
  longitude* string
  id*        integer
}

```

```
}
```

STATUS CODE - 401: Unauthorized - Missing or invalid authentication token.

STATUS CODE - 403: Forbidden - User role not recognized.

STATUS CODE - 404: Not Found - No city exists with the provided ID.

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```
{
  detail [{
    Array of object:
    loc*
      ANY OF
      prop0
        string
      prop1
        integer
    msg*  string
    type* string
    input undefined
    ctx {
    }
  }]
}
```

2.4 GET /city/name/{city_name}

Get City By Name

Get city by name

Fetches a specific city's details using its string name.

Access Level: Admin and User.

Notes:

- The search is case-insensitive as all names are stored in lowercase.

REQUEST

PATH PARAMETERS

NAME	TYPE	DESCRIPTION
*city_name	string	

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
{
  name*    string
  latitude* string
  longitude* string
  id*      integer
}
```



```
}
```

STATUS CODE - 401: Unauthorized - Missing or invalid authentication token.

STATUS CODE - 403: Forbidden - User role not recognized.

STATUS CODE - 404: Not Found - No city exists with the provided name.

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```
{
  detail [{
    Array of object:
    loc*
      ANY OF
      prop0
        string
      prop1
        integer
    msg* string
    type* string
    input undefined
    ctx {
    }
  }]
}
```

2.5 PUT /city/{city_id}

Update City

Update city

Modifies an existing city record based on its ID.

Access Level: Admin only.

Notes:

- All fields in the request body must be provided for a complete update.

REQUEST

PATH PARAMETERS

NAME	TYPE	DESCRIPTION
*city_id	integer	

REQUEST BODY - application/json

```
{
  name* string
  latitude*
    ANY OF
    prop0
      number
      between -90 and 90
    prop1
      string
}
```

```

    PATTERN:^(?!^[ -+.*$])[+-]?0*\d*\.\d*$
longitude*
  ANY OF
  prop0
  number
  between -180 and 180
  prop1
  string
  PATTERN:^(?!^[ -+.*$])[+-]?0*\d*\.\d*$
}

```

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```

{
  name*      string
  latitude*   string
  longitude*  string
  id*         integer
}

```

STATUS CODE - 401: Unauthorized - Missing or invalid authentication token.

STATUS CODE - 403: Forbidden - User does not have administrative privileges.

STATUS CODE - 404: Not Found - The city to update was not found.

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```

{
  detail [{
    Array of object:
    loc*
      ANY OF
      prop0
      string
      prop1
      integer
    msg*  string
    type* string
    input undefined
    ctx {
    }
  }]
}

```

3. JOURNEYS

A collection of Travel Routes which makes a journey between any two given cities. This will return the cheapest possible combination of Travel Routes to complete a journey, as well as cost breakdown.

3.1 GET /journeys/id

Get Journey By Id

Fetch a journey by city IDs

Calculates the cheapest route between two cities using their unique integer identifiers.

The algorithm evaluates multiple transport modes (Rail, FlixBus, National Express) to find the most cost-effective path.

Access Level: Admin and User.

Notes:

- The total_price includes a £3.00 changeover fee for every station transfer required.
- Result includes a full breakdown of ticket costs, applied discounts, and individual journey legs.

REQUEST

QUERY PARAMETERS

NAME	TYPE	DESCRIPTION
*origin_id	integer	
*destination_id	integer	
railcard_discount	integer	
advanced_fares	boolean	
max_coach_legs	integer	

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
{
  total_price*      string
  ticket_price*     string
  railcard_discounts* string
  advanced_discounts* string
  transfer_price*   string
  path* [{
    Array of object:
      price*      string
      notes
      ANY OF
      prop0
      string
      prop1
      null
      id*         integer
      origin_station* {
```

```

    name*      string
    city_id*   integer
    city* {
        name*      string
        latitude*  string
        longitude* string
        id*        integer
    }
}
destination_station* {
    name*      string
    city_id*   integer
    city* {
        name*      string
        latitude*  string
        longitude* string
        id*        integer
    }
}
transport_mode* {
    name* string
    id*   integer
}
}
}]
}

```

STATUS CODE - 401: Unauthorized - Missing or invalid authentication token.

STATUS CODE - 403: Forbidden - User role not recognized.

STATUS CODE - 404: Not Found - No journey could be calculated between the provided IDs.

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```

{
  detail [{
    Array of object:
    loc*
    ANY OF
    prop0
    string
    prop1
    integer
    msg* string
    type* string
    input undefined
    ctx {
    }
  }]
}

```

3.2 GET /journeys/name

Get Journey By Name

Fetch a journey by city names

Calculates the cheapest route between two cities using their string names.

Access Level: Admin and User.

Notes:

- City names are case-insensitive.
- The calculation accounts for railcard discounts and advanced fare estimates if enabled.
- A £3.00 transfer fee is added to the total price for each leg transition.

REQUEST

QUERY PARAMETERS

NAME	TYPE	DESCRIPTION
*origin_name	string	
*destination_name	string	
railcard_discount	integer	
advanced_fares	boolean	
max_coach_legs	integer	

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
{
  total_price*      string
  ticket_price*     string
  railcard_discounts* string
  advanced_discounts* string
  transfer_price*   string
  path* [{
    Array of object:
      price*      string
      notes
      ANY OF
      prop0
      string
      prop1
      null
      id*         integer
      origin_station* {
        name*      string
        city_id*   integer
        city* {
          name*      string
          latitude*  string
          longitude* string
          id*         integer
        }
      }
    }
  }
  destination_station* {
    name*      string
    city_id*   integer
    city* {
      name*      string
```

```

        latitude*  string
        longitude* string
        id*        integer
    }
}
transport_mode* {
    name*  string
    id*    integer
}
}]
}

```

STATUS CODE - 401: Unauthorized - Missing or invalid authentication token.

STATUS CODE - 403: Forbidden - User role not recognized.

STATUS CODE - 404: Not Found - One or both city names do not exist, or no path was found.

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```

{
  detail [{
    Array of object:
    loc*
    ANY OF
    prop0
    string
    prop1
    integer
    msg*  string
    type* string
    input undefined
    ctx {
    }
  }]
}

```

4. STATIONS

4.1 GET /stations/

Get All Stations

Get all stations

Retrieves a complete list of all transport hubs (Rail stations and Coach stops) stored in the database.

Access Level: Admin and User.

REQUEST

No request parameters

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
[ {  
  Array of object:  
    name*      string  
    city_id*   integer  
    city* {  
      name*      string  
      latitude*  string  
      longitude* string  
      id*        integer  
    }  
  }  
}]
```

STATUS CODE - 401: Unauthorized - Missing or invalid authentication token.

STATUS CODE - 403: Forbidden - User role not recognized.

4.2 POST /stations/

Create New Station

Create a new station

Adds a new station record to the database. The system verifies that the associated city exists before the station is created.

Access Level: Admin only.

Notes:

- Each station must be linked to a valid city_id.
- This is a prerequisite for creating routes between different locations.

REQUEST

REQUEST BODY - application/json

```
{  
  name*      string  
  city_id*   integer  
}
```

RESPONSE

STATUS CODE - 201: Successful Response

RESPONSE MODEL - application/json

```
{
  name*      string
  city_id*   integer
  city* {
    name*     string
    latitude*  string
    longitude* string
    id*       integer
  }
}
```

STATUS CODE - 401: Unauthorized - Missing or invalid authentication token.

STATUS CODE - 403: Forbidden - User does not have administrative privileges.

STATUS CODE - 404: Not Found - The provided city_id does not match any existing city.

STATUS CODE - 422: Validation Error - Improperly formatted request body.

4.3 GET /stations/{station_id}

Get Station By Id

Get station by ID

Fetches the details of a specific station, including its parent city information, using its unique integer ID.

Access Level: Admin and User.

REQUEST

PATH PARAMETERS

NAME	TYPE	DESCRIPTION
*station_id	integer	

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
{
  name*      string
  city_id*   integer
  city* {
    name*     string
    latitude*  string
    longitude* string
    id*       integer
  }
}
```

STATUS CODE - 401: Unauthorized - Missing or invalid authentication token.

STATUS CODE - 403: Forbidden - User role not recognized.

STATUS CODE - 404: Not Found - No station exists with the provided ID.

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```
{
  detail [{
    Array of object:
    loc*
    ANY OF
    prop0
    string
    prop1
    integer
    msg* string
    type* string
    input undefined
    ctx {
    }
  }]
}
```

5. TRANSPORT MODES

Categorises routes based on transport mode such as 'train' or 'coach'.

5.1 GET /transport_mode/

Get All Transport Mode

Retrieve all transport modes

Fetches a complete list of all transport categories available in the database.

Access Level: Admin and User.

REQUEST

No request parameters

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
[ {  
  Array of object:  
    name* string  
    id*   integer  
  } ]
```

STATUS CODE - 401: Unauthorized - Missing or invalid authentication token.

STATUS CODE - 403: Forbidden - User role not recognized.

5.2 POST /transport_mode/

Create New Transport Mode

Register a new transport mode

Adds a new category of transport (e.g., Train, Coach) to the system.

Access Level: Admin only.

REQUEST

REQUEST BODY - application/json

```
{  
  name* string  
}
```

RESPONSE

STATUS CODE - 201: Successful Response

RESPONSE MODEL - application/json

```
{  
  name* string  
  id*   integer  
}
```

STATUS CODE - 401: Unauthorized - Missing or invalid authentication token.

STATUS CODE - 403: Forbidden - User does not have administrative privileges.

STATUS CODE - 422: Validation Error - Improperly formatted request body.

5.3 GET /transport_mode/{transport_mode_id}

Get Transport Mode By Id

Get transport mode by ID

Retrieves the specific details of a transport category using its unique integer ID.

Access Level: Admin and User.

REQUEST

PATH PARAMETERS

NAME	TYPE	DESCRIPTION
*transport_mode_id	integer	

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
{
  name* string
  id* integer
}
```

STATUS CODE - 401: Unauthorized - Missing or invalid authentication token.

STATUS CODE - 403: Forbidden - User role not recognized.

STATUS CODE - 404: Not Found - No transport mode exists with the provided ID.

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```
{
  detail [{
    Array of object:
    loc*
    ANY OF
    prop0
    string
    prop1
    integer
    msg* string
    type* string
    input undefined
    ctx {
    }
  }]
}
```

6. TRAVEL ROUTES

The individual intercity legs which make up a Journey.

6.1 GET /routes/

Get All Routes

List all travel routes

Retrieves a complete list of every available route between stations.

Access Level: Admin and User.

Notes:

- Caution: This dataset contains approximately 20,000 records. Requesting this via a web browser may cause significant performance lag or crashes.

REQUEST

No request parameters

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
[{  
  Array of object:  
  price*      string  
  notes  
  ANY OF  
  prop0  
  string  
  prop1  
  null  
  id*         integer  
  origin_station* {  
    name*      string  
    city_id*   integer  
    city* {  
      name*     string  
      latitude* string  
      longitude* string  
      id*       integer  
    }  
  }  
  destination_station* {  
    name*      string  
    city_id*   integer  
    city* {  
      name*     string  
      latitude* string  
      longitude* string  
      id*       integer  
    }  
  }  
}]
```

```

    transport_mode* {
      name* string
      id* integer
    }
  }
}]

```

STATUS CODE - 401: Unauthorized - Missing or invalid authentication token.

STATUS CODE - 403: Forbidden - User role not recognized.

6.2 POST /routes/

Create New Route

Create a new travel route

Adds a new route record to the network. The system validates that the origin station, destination station, and transport mode all exist before saving.

Access Level: Admin only.

REQUEST

REQUEST BODY - application/json

```

{
  price*
    ANY OF
    prop0
    number
    prop1
    string
    PATTERN:^(?!^[-+.*$])[+-]?0*(?:\d{0,8}|(?:=[\d.]{1,11}0*$)\d{0,8}\.\d{0,2}0*$)
  notes
    ANY OF
    prop0
    string
    prop1
    null
  origin_station_id*      integer
  destination_station_id* integer
  transport_mode_id*      integer
}

```

RESPONSE

STATUS CODE - 201: Successful Response

RESPONSE MODEL - application/json

```

{
  price*      string
  notes
    ANY OF
    prop0
    string
    prop1
    null
  id*          integer
  origin_station* {
    name*      string

```

```

    city_id*    integer
    city* {
        name*    string
        latitude* string
        longitude* string
        id*       integer
    }
}
destination_station* {
    name*    string
    city_id* integer
    city* {
        name*    string
        latitude* string
        longitude* string
        id*       integer
    }
}
transport_mode* {
    name* string
    id*   integer
}
}

```

STATUS CODE - 401: Unauthorized - Missing or invalid authentication token.

STATUS CODE - 403: Forbidden - User does not have administrative privileges.

STATUS CODE - 404: Not Found - The origin station, destination station, or transport mode ID does not exist.

STATUS CODE - 422: Validation Error - Improperly formatted request body.

6.3 GET /routes/{route_id}

Get Route By Id

Get specific route details

Fetches the full details of a single travel route, including its pricing and associated station IDs, using its unique identifier.

Access Level: Admin and User.

REQUEST

PATH PARAMETERS

NAME	TYPE	DESCRIPTION
*route_id	integer	

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```

{
  price*    string
  notes
  ANY OF

```

```

    prop0
    string
    prop1
    null
    id*      integer
    origin_station* {
        name*      string
        city_id*    integer
        city* {
            name*      string
            latitude*  string
            longitude* string
            id*        integer
        }
    }
    destination_station* {
        name*      string
        city_id*    integer
        city* {
            name*      string
            latitude*  string
            longitude* string
            id*        integer
        }
    }
    transport_mode* {
        name* string
        id*  integer
    }
}

```

STATUS CODE - 401: Unauthorized - Missing or invalid authentication token.

STATUS CODE - 403: Forbidden - User role not recognized.

STATUS CODE - 404: Not Found - No route exists with the provided ID.

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```

{
  detail [{
    Array of object:
    loc*
    ANY OF
    prop0
    string
    prop1
    integer
    msg* string
    type* string
    input undefined
    ctx {
    }
  }]
}

```

6.4 PUT /routes/{route_id}

Update Route

Update an entire route

Replaces an existing route's data. The system performs a full validation check on station IDs and transport mode IDs before applying the update.

Access Level: Admin only.

Notes:

- All fields must be included in the request body for the update to process.

REQUEST

PATH PARAMETERS

NAME	TYPE	DESCRIPTION
*route_id	integer	

REQUEST BODY - application/json

```
{
  price*
    ANY OF
    prop0
      number
    prop1
      string
    PATTERN:^(?!^[ -+.*$][+-]?0*(?:\d{0,8}|(?:=[\d.]{1,11}0*$)\d{0,8}\.\d{0,2}0*$)
  notes
    ANY OF
    prop0
      string
    prop1
      null
  origin_station_id*      integer
  destination_station_id* integer
  transport_mode_id*      integer
}
```

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
{
  price*      string
  notes
    ANY OF
    prop0
      string
    prop1
      null
  id*         integer
  origin_station* {
    name*      string
    city_id*   integer
  }
```



```

    city* {
      name*      string
      latitude*   string
      longitude*  string
      id*         integer
    }
  }
  destination_station* {
    name*      string
    city_id*   integer
    city* {
      name*      string
      latitude*   string
      longitude*  string
      id*         integer
    }
  }
  transport_mode* {
    name* string
    id*   integer
  }
}

```

STATUS CODE - 401: Unauthorized - Missing or invalid authentication token.

STATUS CODE - 403: Forbidden - User does not have administrative privileges.

STATUS CODE - 404: Not Found - The route, station, or transport mode was not found.

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```

{
  detail [{
    Array of object:
    loc*
    ANY OF
    prop0
    string
    prop1
    integer
    msg* string
    type* string
    input undefined
    ctx {
    }
  }]
}

```

6.5 DELETE /routes/{route_id}

Delete Route

Delete a route

Permanently removes a specific route record from the database.

Access Level: Admin only.

Notes:

- This action is destructive and cannot be undone.
- Associated station or city records are not affected by this operation.

REQUEST

PATH PARAMETERS

NAME	TYPE	DESCRIPTION
*route_id	integer	

RESPONSE

STATUS CODE - 204: Successful Response

STATUS CODE - 401: Unauthorized - Missing or invalid authentication token.

STATUS CODE - 403: Forbidden - User does not have administrative privileges.

STATUS CODE - 404: Not Found - No route exists with the provided ID.

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```
{
  detail [{
    Array of object:
    loc*
      ANY OF
      prop0
        string
      prop1
        integer
    msg* string
    type* string
    input undefined
    ctx {
    }
  }]
}
```

